

Depu Meng, Ph. D.

Senior Research Engineer and Tech Lead at Applied Intuition

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Research Interests

World foundation model: Robotics world model, video generation, world action model, multi-modal foundation model pretraining, end-to-end driving

Education

Microsoft Research Asia - University of Science and Technology of China	BEIJING, CHINA
Ph. D. in Control Science and Engineering	<i>Sept. '18 – Jun. '23</i>
• Advisors: Dr. Baining Guo (Microsoft), Prof. Houqiang Li (USTC)	
University of Science and Technology of China	HEFEI, CHINA
B.E. in Electrical Engineering (School of Gifted Young)	<i>Sept. '14 – Jun. '18</i>

Work Experience

Applied Intuition, Research Team	SUNNYVALE, CA
Senior Research Engineer and Tech Lead	<i>Sep. '25 – present</i>
• Leading world foundation model and and physical world model research.	
DiDi Autonomous Driving, AI Research Team	SAN JOSE, CA
Machine Learning Engineer	<i>Dec. '24 – Aug. '25</i>
• End-to-end foundation model training and 3D Gaussian Splatting.	
University of Michigan	ANN ARBOR, MI
Post-Doctoral Research Fellow	<i>Aug. '23 – Dec. '24</i>
• Multi-modal traffic-safety foundation model development.	
• Led the team won USDOT Intersection Safety Challenge Stage 1A and 1B.	

Internship Experience

University of Michigan	ANN ARBOR, MI
Research Assistant	<i>Apr. '22 – Aug. '23</i>
• Image generation and driving scenario simulation.	
Mentor: Prof. Henry X. Liu	
Meituan Autonomous Delivery Group	BEIJING, CHINA
Research Intern	<i>Aug. '21 – Apr. '22</i>
• Joint Lidar-based object detection and motion prediction.	
Mentor: Dr. Changqian Yu	
Microsoft Research Asia, Visual Computing Group	BEIJING, CHINA
Research Intern	<i>Jul. '19 – Jul. '21</i>
• Transformer-based object detection and unsupervised pretraining.	
Mentor: Dr. Jingdong Wang	
Microsoft Research Asia, Visual Computing Group	BEIJING, CHINA
Research Intern	<i>Jul. '17 – Jul. '18</i>
• CNN architecture design and image classification.	
Mentor: Dr. Jingdong Wang	

Publications

* equal contribution, † project lead.

Teaching Video Generators to Remember: Eliciting Dynamic Memory for Out-of-Sight State Evolution

Tianshuo Xu, Yichen Xie, **Depu Meng**[†], Chensheng Peng, Quentin Herau, Bo Jiang, Yihan Hu, Wei Zhan

arXiv preprint, 2026

[\[PAPER\]](#) [\[PROJECT\]](#) [\[CODE\]](#) [\[MODEL\]](#) [\[DATASET\]](#)

LaMo: Self-Supervised Latent Motion Priors for Physical Realism in Video Generation

Bo Jiang, **Depu Meng**, Yihan Hu, Yichen Xie, Tianshuo Xu, Wei Zhan

arXiv preprint, 2026

[\[PAPER\]](#) [\[PROJECT\]](#) [\[CODE\]](#)

SpectralSplat: Appearance-Disentangled Feed-Forward Gaussian Splatting for Driving Scenes

Quentin Herau*, Tianshuo Xu*, **Depu Meng**[†], Jiezhi Yang, Chensheng Peng, Spencer Sherk, Yihan Hu, Wei Zhan

arXiv preprint, 2026

[\[PAPER\]](#)

URoPE: Universal Relative Position Embedding across Geometric Spaces

Yichen Xie, **Depu Meng**, Chensheng Peng, Yihan Hu, Quentin Herau, Masayoshi Tomizuka, Wei Zhan

ECCV, 2026 (*Oral*)

[\[PAPER\]](#) [\[PROJECT\]](#) [\[CODE\]](#)

Exploring Communication and Roadside Perception Requirements for Cooperative Warning Systems at Intersections

Tinghan Wang, **Depu Meng**, Boqi Li, Rusheng Zhang, Yukun Zuo, Shengyin Shen, Darian Hogue, Michael Maile, Michael Shulman, Henry X. Liu

IEEE IV, 2025

[\[PAPER\]](#)

Towards Comprehensive Roadside Intelligence: Sensor Fusion and Full-Stack Perception with Multiple Cameras

Rusheng Zhang, **Depu Meng**, Shengyin Shen, Boqi Li, Tinghan Wang, Henry X. Liu

IEEE IV, 2025

[\[PAPER\]](#)

MSight: An Edge-Cloud Infrastructure-Based Perception System for Connected Automated Vehicles

Rusheng Zhang*, **Depu Meng***, Shengyin Shen, Zhengxia Zou, Houqiang Li, Henry X. Liu

Technical Report, 2024

[\[PAPER\]](#) [\[CODE\]](#)

Robust Roadside Perception: An Automated Data Synthesis Pipeline Minimizing Human Annotation

Rusheng Zhang, **Depu Meng**, Lance Bassett, Shengyin Shen, Zhengxia Zou, Henry X. Liu

IEEE T-IV, 2024

[\[PAPER\]](#)

Evaluating Roadside Perception for Autonomous Vehicles: Insights from Field Testing

Rusheng Zhang, **Depu Meng**, Shengyin Shen, Tinghan Wang, Tai Karir, Michael Maile, Henry X. Liu

arXiv preprint, 2024

[\[PAPER\]](#)

ROCO: A Roundabout Traffic Conflict Dataset

Depu Meng, Owen Sayer, Rusheng Zhang, Shengyin Shen, Houqiang Li, Henry X. Liu

TRB Annual Meeting, 2023

[\[PAPER\]](#) [\[DATASET\]](#)

HyMo: Hybrid Motion Representation Learning for Prediction from Raw Sensor Data

Depu Meng, Changqian Yu, Jiajun Deng, Deheng Qian, Houqiang Li, Dongchun Ren

IEEE T-MM, 2023

[\[PAPER\]](#)

CORE: Consistent Representation Learning for Face Forgery Detection

Yunsheng Ni, **Depu Meng**, Changqian Yu, Chengbin Quan, Dongchun Ren, Youjian Zhao

CVPR Workshops, 2022

[\[PAPER\]](#) [\[CODE\]](#)

Conditional DETR for Fast Training Convergence

Depu Meng^{*}, Xiaokang Chen^{*}, Zejia Fan, Yuhui Yuan, Gang Zeng, Houqiang Li, Lei Sun, Jingdong Wang

ICCV, 2021

[\[PAPER\]](#) [\[CODE\]](#) [\[MODEL\]](#)

Consistent Instance Classification for Unsupervised Representation Learning

Depu Meng, Zigang Geng, Zhirong Wu, Bin Xiao, Houqiang Li, Jingdong Wang

ICCV Workshops, 2021

[\[PAPER\]](#)

Bottom-Up Human Pose Estimation by Ranking Heatmap-Guided Adaptive Keypoint Estimates

Ke Sun, Zigang Geng, **Depu Meng**, Bin Xiao, Dong Liu, Zhaoxiang Zhang, Jingdong Wang

arXiv preprint, 2020

[\[PAPER\]](#) [\[CODE\]](#)

Deep Convolutional Neural Networks with Merge-and-Run Mappings

Liming Zhao, Mingjie Li, **Depu Meng**, Xi Li, Zhuowen Tu, Zhaoxiang Zhang, Yueting Zhuang, Jingdong Wang

IJCAI, 2018

[\[PAPER\]](#) [\[CODE\]](#)

Awards

U.S. DOT Intersection Safety Challenge Stage 1B Tier 1 Winner Team (Prize: \$750,000)	Jan. '25
U.S. DOT Intersection Safety Challenge Stage 1A Winner Team (Prize: \$100,000)	Jan. '24
Shenzhen Stock Exchange Scholarship, USTC	Dec. '22
Star of Tomorrow Internship Award, Microsoft Research Asia	Jul. '18
First Prize in Intelligent Robot Competition, Harbin Institute of Technology	Jul. '16
The AEGON-INDUSTRIAL Fund Scholarship, USTC	Oct. '15

Services

Conference Reviewer: CVPR, ECCV, ICCV, NeurIPS, ACM MM, TRBAM, IROS

Journal Reviewer: IJCV, IEEE T-IV, IEEE T-MM, IEEE T-CSVT, Neurocomputing, Pattern Recognition